Sets and Functions

October 18, 2026

We do not attempt to define the word *set*. Intuitively a set is a *collection*, or *aggregate*, or *family*, or *ensemble* (all of which words are used synonymously with set) of objects which are called the *elements*, or *members* of the set, and the set is completely determined by the knowledge of which objects are elements of it. We may speak, for example, of the set of students at a certain university; the elements of this set are the individual students there. Similarly we may speak of the set of all real numbers, or the set of all straight lines in a given plane, etc. It should be noted that the elements of a set may themselves be sets; for example, each element of the set of the set of all straight lines in a given plane is a set of points, and we may also consider such less mathematical examples as the set of married couples in a given town, or the set of regiments in an army.

We shall generally use capital letters to denote sets and lower-case letters to denote their elements. The symbol $\in$ is used to denote membership in a set, so that

$$x \in S$$

means that x is an element of the set S. The statement "x is not an element of S" is abbreviated

$$x \notin S.$$

Instead of writing $a \in S$, $b \in S$, $c \in S$ (the commas having the same meaning as "and") we often write $a, b, c \in S$.

Sets are sometimes indicated by listing all their members between braces. For example, the set

$$\{\text{Jane}, \text{Jim}\}$$

has Jane and Jim as its members,

$$\{1, 2, 3, ...\}$$

(with the three dots read "and so on") is the set of positive integers, and

$$\{a\}$$

is the set having one element, the object named a. Note that $\{a\}$ is not the same as a. In the same way there is a difference between a university class consisting of one student and the student himself, or between a committee consisting of one person and that person. The above notation however is not always feasible. A more frequently used notation is

$$\{x : (\text{statement involving } x)\},$$

which means the set of all x for which the statement is true. Thus if $\mathbf{R}$ is the set of real numbers,

$$\{x \in \mathbf{R} : x^2 = 1\} = \{1, -1\}.$$

If X and Y are sets and every element of X is also an element of Y, we say that X is a **subset** of Y; this is written

$$X \subset Y, \quad \text{or} \quad Y \supset X.$$

Thus $X \subset Y$ is shorthand for the statement "if $x \in X$ then $x \in Y$." $X = Y$ is equivalent to the two statements $X \subset Y$ and $Y \subset X$. If $X \subset Y$ and $Y \subset Z$ then $X \subset Z$; the two first statements are sometimes written more succinctly

$$X \subset Y \subset Z.$$

For any object x and set X the relation $x \in X$ can now be written in another (less convenient!) way, namely $\{x\} \subset X$.

The statement $X = Y$ means that $X \subset Y$ and $Y \subset X$.

IMPORTANT: To show that two sets A and B are equal, prove that $A \subset B$ and $B \subset A$.

The **empty set** is the set with no elements. It is denoted by the symbol $\emptyset$. A source of confusion to beginners is that although the empty set contains *nothing*, it itself is *something* (namely some particular set, the one characterized by the fact that nothing is in it). The set $\{\emptyset\}$ is a set containing exactly one element, namely the empty set. In a similar way, when dealing with numbers, say with ordinary integers, we must be careful not to regard the number zero as nothing: zero is *something*, a particular number, which

represents the number of things in "nothing." Thus zero and $\emptyset$ are quite different, but there is a connection between them in that the set $\emptyset$ has zero elements.

Note that for any set X we have

$$\emptyset \subset X \quad \text{and} \quad X \subset X.$$

A special case of both of these statements is the statement

$$\emptyset \subset \emptyset,$$

which occasions difficulty if, as is often improperly done, one reads "is contained in" for both of the symbols $\subset$ and $\in$. The statement $\emptyset \subset \emptyset$ is true because the statement "for each $x \in \emptyset$ we have $x \in \emptyset$" is obviously true, and also because it is "vacuously true," that is there is no $x \in \emptyset$ for which the statement must be verified, just as the statement "all pigs with wings speak Chinese" is vacuously true.

Operations on Sets

If X and Y are sets, the **intersection of X and Y**, denoted by $X \cap Y$, is defined to be the set of all objects which are both elements of X and elements of Y. In symbols,

$$X \cap Y = \{x : x \in X \text{ and } x \in Y\}.$$

The **union of X and Y**, denoted by $X \cup Y$, is the set of all objects which are elements of at least one of the sets X and Y. That is, $X \cup Y$ is the set of all objects which are either elements of X or elements of Y (or of both), in symbols

$$X \cup Y = \{x : x \in X \text{ or } x \in Y\}.$$

The word "or" is used here in the manner that is standard in mathematics. In ordinary language the word "or" is often exclusive, that is, if A and B are statements, then "A or B" is understood to mean "A or B but not both," whereas in mathematics it always means "A or B, or both A and B."

If X is a subset of a set S, then the **complement** of X in S is the set of all elements of S which are not elements of X. If it is explicitly stated, or clear from the context, exactly what the set S is, we often omit the words "in S" and use the notation X^c for the complement of X. That is,

$$X^c = \{x \in S : x \notin X\}.$$

If X and Y are sets, the notation $X - Y$ is sometimes used for $\{x \in X : x \notin Y\}$. Thus if X and Y are subsets of some set S, then $X - Y = X \cap Y^c$.

Functions

If X and Y are sets, by a **function from X to Y** is meant a rule which associates with each element of X a definite element of Y. The "rule" can be given in many ways, but the essential thing is that given any element of X there is associated, somehow, some definite element of Y. Two functions from X to Y are considered equal if and only if both functions associate with each specific element of X the same element of Y.

Exercises

Exercise 1. Let X be a set and let A and B be subsets of X. Prove that $A \subseteq B$ if and only if $B^c \subseteq A^c$.

Solution. Suppose that $A \subseteq B$. Let $x \in B^c$. Then x is not in B. Since $A \subseteq B$, it follows that x is not in A, that is, it is in A^c. Conversely, suppose that $B^c \subseteq A^c$. Let $x \in A$. Then x is not in A^c. Therefore, x is not in B^c, that is, $x \in B$.

Exercise 2. Let $f\colon X \to Y$ be a function, let A and B be subsets of X, and let C and D be subsets of Y. Prove that:

(a) $f(A \cup B) = f(A) \cup f(B)$

(b) $f(A \cap B) \subset f(A) \cap f(B)$

Solution. (a) Let $y \in f(A \cup B)$. Then there is $x \in A \cup B$ such that $y = f(x)$. If $x \in A$, then $y = f(x) \in f(A)$. If $x \in B$, then $y = f(x) \in f(B)$. So $y \in f(A) \cup f(B)$. Conversely, if $y \in f(A) \cup f(B)$, then either $y \in f(A)$ or $y \in f(B)$. If $y \in f(A)$, then there is $x \in A$ such that $y = f(x)$. If $y \in f(B)$, then there is $x \in B$ such that $y = f(x)$. So $y = f(x)$ for some $x \in A \cup B$.

(b) Let $y \in f(A \cap B)$. Then there is $x \in A \cap B$ such that $y = f(x)$. Since $x \in A$, we have $y = f(x) \in f(A)$. Similarly, since $x \in B$, we have $y \in f(B)$. Therefore, $y \in f(A) \cap f(B)$.

Exercise 3. Let $f\colon X \to Y$ be a function, let A and B be subsets of X, and let C and D be subsets of Y. Prove that:

(a) $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$

(b) $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$

Solution. (a) Let $x \in f^{-1}(C \cup D)$. Then $f(x) \in C \cup D$. If $f(x) \in C$, then $x \in f^{-1}(C)$. If $f(x) \in D$, then $x \in f^{-1}(D)$. So $x \in f^{-1}(C) \cup f^{-1}(D)$.

(b) Similar to (a).

Exercise 4. Let $f\colon X \to Y$ be a function, let A and B be subsets of X, and let C and D be subsets of Y. Prove that:

(a) $f^{-1}(f(A)) \supset A$

(b) $f(f^{-1}(C)) \subset C$

Solution. (a) Let $x \in A$. Then $x \in f^{-1}(f(x))$.

(b) Let $y \in f(f^{-1}(C))$. Then there is $x \in f^{-1}(C)$ such that $y = f(x)$. Since $x \in f^{-1}(C)$, we know that $f(x) \in C$. Therefore, $y \in C$.

Exercise 5. (a) How many subsets are there of the set $\{1, 2, 3, ..., n\}$?

(b) How many maps of this set into itself are there?

(c) How many maps of this set onto itself are there?

Solution. (a) 2^n, because each element is either in the set or not in the set.

(b) $n!$

(c) $n^n + \binom{n}{n-1} n^{n-1} + \binom{n}{n-2} n^{n-2} + \cdots + \binom{n}{1} \cdot 1$

Exercise 6. Prove that if $f\colon A \to B$ and $g\colon B \to C$ are bijections, then so is $g \circ f$.

Solution. Let $c \in C$. Then since g is surjective, there is $b \in B$ such that $g(b) = c$. Since f is surjective, there is $a \in A$ such that $b = f(a)$. Therefore, $c = g(f(a)) = (g \circ f)(a)$, and so $g \circ f$ is surjective.

Suppose that $(g \circ f)(a) = (g \circ f)(a')$. Then $g(f(a)) = g(f(a'))$. Since g is injective, $f(a) = f(a')$. Since f is injective, $a = a'$. Therefore, $g \circ f$ is injective.